

武汉大学数学与统计学院 2024-2025 第一学期
抽象代数 期中 ZUOKANG

1. Let p be a prime and r, k be positive integers.
 - (a) Show that $(1+p)^{p^k} \equiv 1 + p^{k+1} \pmod{p^{k+2}}$.
 - (b) Show that for $\bar{2} \in \mathbb{Z}_{5^r}^*$, $o(\bar{2}) = 4 \times 5^{r-1}$ for some integer $0 \leq s \leq r-1$.
 - (c) Show that $\mathbb{Z}_{5^r}^* \cong C_{5^{r-1}} \times C_4$ as groups.
2. Prove that if $G/Z(G)$ is cyclic, then G is abelian.
3.
 - (a) Show that D_4 is isomorphic to a subgroup $H \leq S_4$, and write down two generators of H .
 - (b) Show that S_4 has no normal subgroup of order 8.
4. Let $f : G \rightarrow G$ be a group homomorphism.
 - (a) Define $f^{(1)} \triangleq f : G \rightarrow G$ and $f^{(n+1)} \triangleq f^{(n)} \circ f = f \circ f^{(n)} : G \rightarrow G$. Show that $f^{(n)}$ is a group homomorphism.
 - (b) $\ker f \subset \ker f^{(2)} \subset \dots \subset \ker f^{(n)} \subset \ker f^{(n+1)} \subset \dots$
 - (c) If G has only finitely many subgroups and f is onto, then f is an isomorphism.
5. Let $H \leq G$, where G is a finite group. For $\forall x \in G$, define $x^G \triangleq \{gxg^{-1} \mid g \in G\}$ and $x^H \triangleq \{h x h^{-1} \mid h \in H\}$.
 - (a) Let $C_G(x) \triangleq \{g \in G \mid gx = xg\}$, show that $C_G(x) \leq G$.
 - (b) Show that $|x^G| = [G : C_G(x)]$ and $|x^H| = [H : C_H(x)]$, where $C_H(x) = H \cap C_G(x)$.
 - (c) If $[G : H] = 2$, then either $|x^G| = |x^H|$ or $|x^G| = 2|x^H|$. (Hint: By 2nd isom thm)
6. Let

$$G \triangleq \left\{ A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \mid a, b, c, d \in \mathbb{C}, \det A = 1, \overline{A}^T \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} A = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \right\}$$

- (a) Show that G is a group under matrix multiplication. And $\forall A \in G$,

$$A = \begin{pmatrix} a & b \\ \bar{b} & \bar{a} \end{pmatrix} \quad \text{with } a\bar{a} - b\bar{b} = 1.$$

- (b) Show that $f : G \rightarrow SL_2(\mathbb{R})$ defined by $A \mapsto X^{-1}AX$, where $X = \begin{pmatrix} 1 & -i \\ 1 & i \end{pmatrix}$, is a group isomorphism.

- 7.** Let $C([0, 1])$ be the set of $\mathbb{R}$ -valued continuous functions over $[0, 1]$.

- (a) Show that $C([0, 1])$ is a ring under function addition and multiplication.
- (b) Show that for any $f \in C([0, 1])$, f is a unit if and only if f is nonzero everywhere.
- (c) For $x \in [0, 1]$, show that $m_x := \{f \in C([0, 1]) \mid f(x) = 0\}$ is an ideal, and it is a maximal ideal.

- 8.** Let $\mathbb{Z}[\sqrt{3}] = \{n + m\sqrt{3} \mid n, m \in \mathbb{Z}\}$.

- (a) Show that $(\sqrt{3})$ is a prime ideal.
- (b) Show that (3) is not a prime ideal.
- (c) There exist infinitely many units in $\mathbb{Z}[\sqrt{3}]$.

- 9.** Let $S \subseteq R$ be nonzero rings, $I \subset R$ be an ideal and $R/I \cong \mathbb{Z}/n\mathbb{Z}$. Prove that $R = S + I$.

- 10.** Let p be a prime number. $\mathbb{Z}_{(p)} = \left\{ \frac{n}{m} \in \mathbb{Q} \mid p \text{ does not divide } m \right\}$.

- (a) Let R be a ring. Show that any non-trivial ring homomorphism $f : \mathbb{Z}_{(p)} \rightarrow R$ satisfies $f(n) = n \cdot f(1) = n \cdot 1_R$.
- (b) Find a non-trivial ring homomorphism $\varphi : \mathbb{Z}_{(p)} \rightarrow \mathbb{Z}/p\mathbb{Z}$.
- (c) Determine $\ker(\varphi)$ explicitly.